

# Electrical Topology Diagram (Implementation v6)

As-of date: 2026-08-12

Purpose: provide a complete, install-level electrical topology for the current build scope, including all major electrical components, fuse IDs, fuse housings, planned wire gauges, and estimated one-way run lengths for procurement planning.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Detailed fuse matrix: docs/implementation/ELECTRICAL\_fuse\_schedule.md
- Battery and trunk recalculation record: docs/studies/ELECTRICAL\_battery\_fuse\_wire\_recalc\_2026-02-18.md
- Decisions and unresolved items: docs/core/TRACKING.md
- Procurement source of truth: bom/bom\_estimated\_items.csv

## Sweep Outcomes Included In This Revision

- Keeps alternator charging architecture on the dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 baseline); obsolete pre-Mechman charger paths are removed from primary topology.
- Keeps Lynx Slot 3 branch to alternator input with F-04 200A/80V MEGA.
- Removes obsolete pre-Mechman engine-bay fuse/conductor placeholders from active architecture.
- Clarifies the confirmed PH-VAN harness as one 15A high-interrupt in-line-fused combined power/positive-sense red lead ( F-12/F-13-PHVAN ) and locks the unfused purple/grey pair to the Wakespeed shunt on the battery side of the hard-mounted SmartShunt in the common battery-negative path.
- Adds Ford Upfitter #3 -> F-15 -> WS500 brown ignition manual alternator-control path.
- Adds the detailed Mechman/WS500/APM-48 install guide as the shop reference for staged installation, first-run checks, and load-dump/shutdown handling.
- Defines APM-48 as a parallel surge clamp at the alternator rather than a series charge-current device.
- Added explicit fuse-holder/housing definitions for every fuse family ( Class T , Lynx MEGA , inline MIDI/ANL/AMI , PV gPV , and AT0/ATC ).
- Added conductor schedule across 48V , 12V , PV, and AC segments with explicit assumptions.
- Updated 12V topology to a shared 12V junction fed by an Orion-Tr Smart 48/12-30 charger and a 12V 100Ah buffer battery branch, with F-11 source fuse plus SW-12V-BATT manual isolation.
- Added a full-circuit estimated run-length validation pass ( C-01 through C-40 ) and purchase-ready wire rollup totals.

## Current Commissioning Snapshot ( 2026-09-02 )

- 48V bus live-tested at 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II inverter mode tested with inverter light on, slight normal hum, and no reported errors.
- SmartShunt and Orion-Tr Smart connected in VictronConnect.
- Cerbo GX access point/remote-console workflow active; Cerbo is powered from a small inline fused 48V feed and connected to MultiPlus via VE.Bus RJ45.
- Short AC-in shore-charge test passed at household-source current limits: about 1294W shore input and about 54.3V x 21.6A battery charging in bulk.
- MultiPlus LiFePO4 charge profile is programmed/owner-verified by supervised first-battery behavior. Owner reports the 3x 48V bank has been permanently paralleled and the 12V , 48V , and 120V systems have been used regularly. Hold open only the explicit protective-earth/bonding and GFCI/polarity checks, alternator commissioning, any unfinished Cerbo/service-face mounting, and final strain-relief/abrasion-control.

## Current Physical Installation Snapshot ( 2026-09-02 )

- The electrical module is hard-mounted through the finished Lonseal/plywood floor into registered truck-bed hardpoints and tied into the hard-mounted Bench/Galley extrusion structure. Owner reports the batteries and ICECO are positively restrained; remaining mobile retention is the not-yet-installed air fryer, bench lid, monitor/laptop hardware, other storage/cargo, and any unfinished terminal/cable protection.
- The driver-rear shore inlet and exterior-to-module cable route are installed. All 120V work, including the four duplex outlet boxes, is owner-reported enclosed and in regular use. Preserve explicit PE/chassis/shell bonding, neutral-isolation, polarity, and GFCI checks.
- All three batteries' positive and negative 2/0 AWG branch cables are complete and the bank has been permanently paralleled for an extended period.
- The 12V buffer-battery/Orion branch is owner-reported operational. Keep its negative direct to the fuse-panel main negative stud, not through SW-12V-BATT ; the positive path remains F-11 -> SW-12V-BATT -> panel main + .

- Owner report 2026-09-02 : Starlink Standard 4 X, ICECO/fridge, water pump, and the installed 12V charging outlets have seen regular use. Starlink uses a dedicated 20A blade fuse; the three separately fused charging outlets are driver-side 100W and two passenger-side 65W . Preserve exact slot/fuse labels, polarity, terminal covers, loaded voltage-drop/heat checks, and final switch/service-face mounting without reopening their functional status.
- Owner report 2026-09-02 : C-31 Desk/office and C-32 Galley/passenger branches each feed one origin GFCI plus one downstream duplex, for four duplex devices/eight plug positions total. Their boxes/enclosures and regular-use state are closed; retain explicit proof of breaker values, branch separation, LINE/LOAD/PE , downstream protection, polarity, and trip/reset in inverter and accepted-shore modes.
- The current narrow switch bundle is temporary and not acceptable as a finish state. De-energize correctly, individually insulate/label DC switch terminals, and mount them on a removable plywood/ABS service face. Any adjacent 120V receptacle stays in its own listed covered box with cable clamps and PE continuity; do not mount exposed AC device backs in the DC switch cavity.

## Orion fuse discriminator — one input fuse only

- Lynx Slot 4 / F-05 : **Orion 48V input positive**, one verified 40A MEGA ( 58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) feeding the existing 6 AWG directly. This is the final single input fuse.
- F-06 : **retired standalone Orion input-fuse position**. Do not install MIDI, FKS/ATO, DIN, or a second fuse after Slot 4.
- Lynx Slot 2 / F-03 : **MPPT branch**, 60A/80V MEGA. If the physically X-marked/misrated Lynx fuse is in Slot 2, replace it with the purchased Victron 60A/80V MEGA; it is not the Orion input fuse.
- F-07 : **Orion 12V output positive**, 60A/80V MEGA in the separate Victron holder near the Orion.
- F-11 : **12V buffer-battery positive**, 100A ANL near the battery, upstream of SW-12V-BATT .
- No 32V -rated fuse is acceptable on any energized 48V branch. Identify the physical X-marked fuse by Lynx slot before replacing or discarding it.

## Length Estimation Defaults Used In This Pass

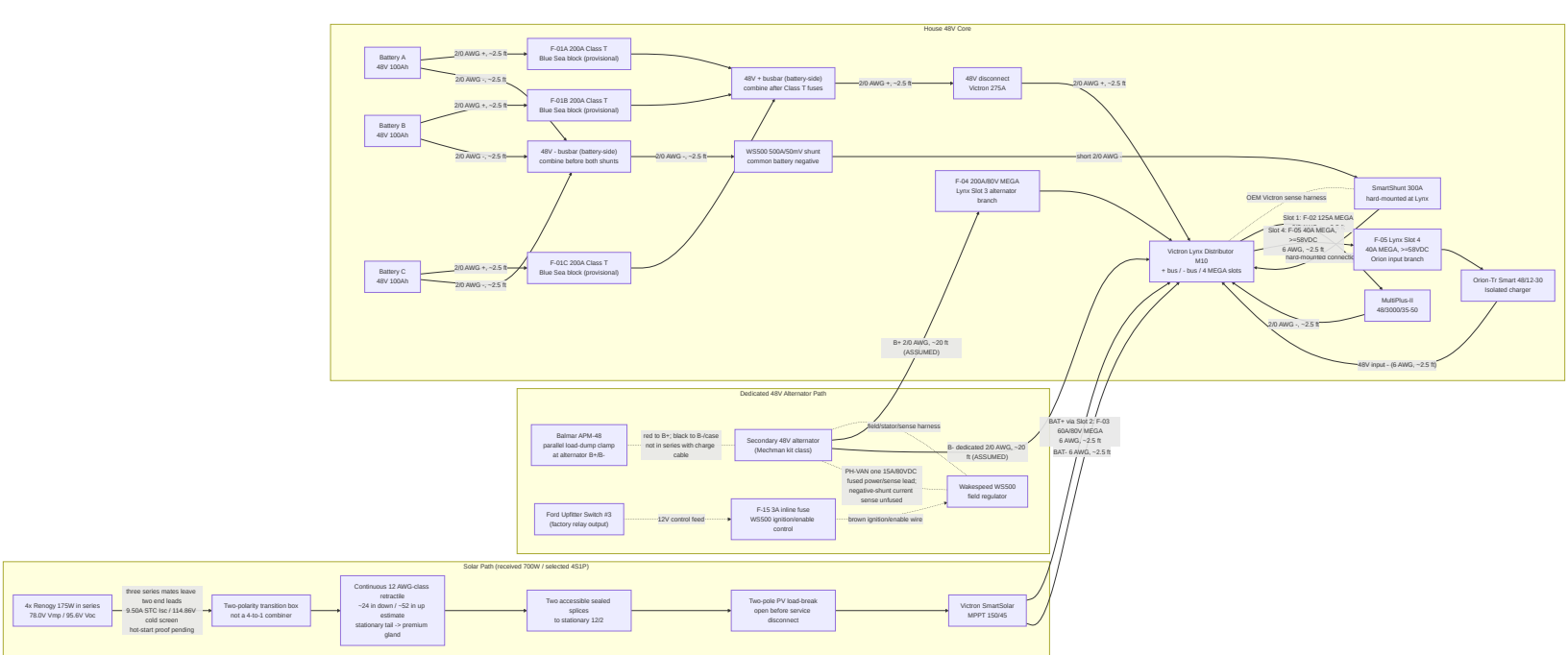
1. Cabinet internal interconnect default: 2.5 ft one-way ( ASSUMED ).
2. Cabinet-to-near load branch default: 8 ft one-way ( ASSUMED ).
3. Cabinet-to-far load branch default: 12 ft one-way ( ASSUMED ).
4. AC branch to receptacle chain default: 15 ft one-way per branch leg ( ASSUMED ).
5. Policy lock: use the smallest gauge that meets current and voltage-drop targets; do not auto-upsize, but flag warnings when margin is tight.
6. Parallel battery bank lock: keep each battery's **total positive + negative path resistance** similar. Equal positive-only length is not required if the positive-length differences are offset by negative-length differences; do not add unnecessary 2/0 cable coils solely for cosmetic equality.

## Battery Fuse/Wire Recalculation Basis (2026-02-18 + 2026-03-19 sync)

- Scope in this pass is limited to battery-side and major 48V trunk paths ( C-01 through C-15 ).
- Provisional battery listing inputs used: 51.2V 100Ah ,  $\leq 200A$  current limit per battery.
- Conservative sizing factors used in this pass:
  1. Parallel-sharing factor  $K_{share} = 1.5$
  2. Continuous margin factor  $K_{cont} = 1.25$
- Current envelope used for battery-discharge branch sizing:  $I_{total} = F-02 + F-05 = 125A + 40A = 165A$  .
- Per-battery design current:  $I_{batt\_design} = (165A / 3) * 1.5 = 82.5A$  .
- Continuous-adjusted minimum battery branch fuse threshold:  $82.5A * 1.25 = 103.1A$  .
- Provisional battery branch fuse selection: F-01A/B/C = 200A Class T , constrained by the provisional battery  $\leq 200A$  current-limit listing.
- Final lock gate: validate true 51.2V battery datasheet/manual current and terminal limits before permanent fuse lock; if lower limits are confirmed, move to 175A .
- Cable procurement remains estimate-based until CAD/field run lengths are frozen. This pass sets a no-padding 2/0 estimate baseline of 77.5 ft total ( 42.5 ft red, 35.0 ft black), including the dedicated alternator migration path.

## Complete Power Topology (48V Core + Charge Sources)

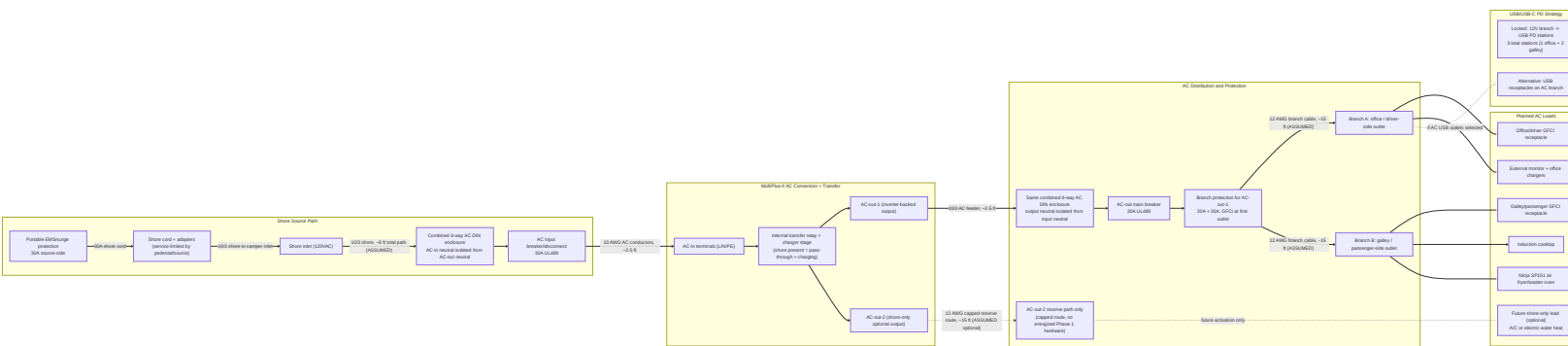
# 12V Distribution Topology (Shared Junction With Buffer Battery)



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- The diagram illustrates a 12V DC power distribution system. At the top, a 12V 100Ah LiFePO4 buffer battery is connected to a 100A class battery main fuse via a 4 AWG, ~2.5 ft to main + stud. The main fuse is connected to a 12V fuse block, which contains a 12-circuit integrated negative bus and main + / - studs (source combine + shared return). The Orion-Tr Smart 48/12-30 isolated charger is connected to the main + stud and the negative bus via 6 AWG, ~2.5 ft to main + stud and 6 AWG, ~2.5 ft to main - / neg bus. The 12V fuse block is connected to various loads, including a 12V-01 Starlink Standard 4 X direct DC 20A blade, a 12V-02 Fridge, a 12V-03 Diesel heater, a 12V-04 Water pump, a 12V-05 CO+Propane detector, a 12V-06 LED lights + dimmer, a 12V-08 Office USB PD station, a 12V-09 Galley USB PD station A, a 12V slot to verify Galley USB PD station B, a 12V-10 Maxxair fan, a 12V-11 DC ambient/cabinet LED strips, a 12V-12 Kicker KMC2 media center, and a 12V-AUDIO-SUB Kicker PTRTP10 powered sub. The diagram also shows a maintenance charge path to the buffer battery and a manual battery disconnect (SW-12V-BATT) connected to the main + stud.

- Hiatus pre-installed 12V branches are tracked here as 12V-06 (factory LED lights with dimmer) and 12V-10 (Maxxair fan); verify final installed branch labeling during electrical audit.
- Planned Govee/ambient strip lighting is a separate branch ( 12V-11 ) and not part of the Hiatus factory 12V-06 lighting circuit.
- DC-first lighting intent: keep ambient/cabinet strip lighting on 12V-11 so nighttime lighting does not require inverter operation.
- Camper audio intent: Kicker 46KMC2 source unit uses a 15A 12V-12 branch from the fuse block, while Kicker 49PTRTP10 powered sub uses a separate 40A fused 4 AWG branch from the 12V source/junction rather than a normal blade-fuse panel branch. Bass peaks may exceed the Orion 30A continuous feed and are supported by the 12V buffer battery for short durations.

## AC Path Topology (Shore + Inverter Output, Full Hierarchy)



## AC Operating Behavior (Expected)

- Shore present: MultiPlus transfer relay closes, AC-in is passed to AC-out paths, and charger stage charges the 48V bank.
- Shore absent: MultiPlus transfers to inverter mode and powers AC-out-1 from battery; AC-out-2 drops by design.
- AC-in hardware is 30A ( source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> 30A breaker -> 10 AWG AC-in conductors ); set the MultiPlus limit to source-specific permitted draw— 10A for first household tests and 12A maximum policy on a normal 15A outlet, or the accepted policy/rating for 20A / 30A service.
- Initial battery charging may use AC-in-only mode with AC-out branch loads disconnected.

## AC Safety/Protection Chain (What Must Exist)

- Upstream shore protection chain before MultiPlus AC-in: shore source/adapters -> portable EMS -> shore cord -> shore inlet -> 30A AC input breaker/disconnect.
- AC-out protection chain: MultiPlus AC-out-1 -> 10/3 feeder -> 30A AC-out main breaker -> 20A branch breakers -> GFCI receptacles.
- Single-enclosure DIN architecture with isolated AC-in and AC-out neutral paths and no neutral mixing.
- Continuous equipment grounding path from shore inlet through MultiPlus and branch circuits. The MultiPlus external M6 PE lug gets at least 4 mm<sup>2</sup>, selected as 10 AWG green stranded copper, to a verified truck-chassis bond point; the aluminum shell gets a separate corrosion-compatible bond into the same equipment-ground network and is not the only PE path.
- Neutral/ground handling follows the MultiPlus internal relay: AC-out neutral bonds to chassis in inverter mode and opens that bond when shore is accepted. Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.
- Keep PE and DC return roles separate: no MultiPlus case/PE jumper to Lynx or 12V negative. Confirm whether the Mechman is isolated- or case-grounded and prove no chassis path bypasses the SmartShunt before alternator commissioning.

## AC Reference Basis (Manufacturer Guidance)

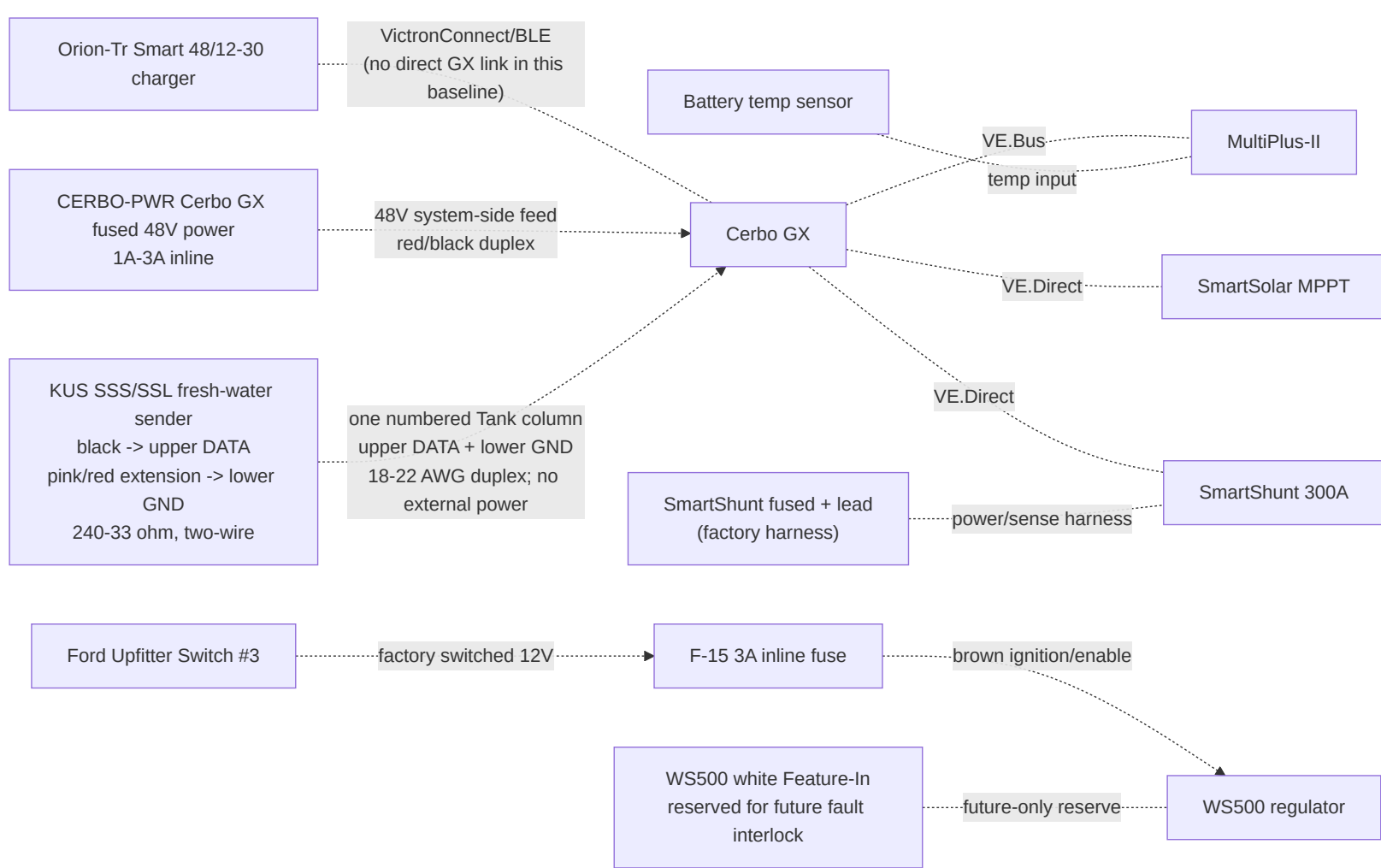
- Victron MultiPlus-II 120V installation guidance ( AC-in breaker sizing, UL943-class residual-current protection on outputs, and AC-out-2 shore-only behavior): [https://www.victronenergy.com/media/pg/MultiPlus-II\\_120V/en/installation.html](https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html)
- Victron MultiPlus-II datasheet ( 48/3000/35-50 baseline model reference): <https://www.victronenergy.com/upload/documents/Datasheet-MultiPlus-II-inverter-charger-120V-EN.pdf>

## AC/USB Baseline Locked For BOM

- Shore interface: 30A RV-style inlet baseline with adapter kit for 15A / 20A hookups.
- AC input protection: portable EMS + combined DIN enclosure + 30A AC input breaker/disconnect upstream of MultiPlus AC-in.
- AC-out-1 distribution: 30A AC-out main plus two protected 20A branches with GFCI-at-first-outlet strategy.
- Receptacle plan/as-built report ( 2026-08-27 ): Branch A office/driver and Branch B galley/passenger each have one first-in-chain 20A self-test GFCI plus one downstream duplex on the corresponding GFCI LOAD , for four duplex devices/eight plug positions total. Keep both 20A branches, 12 AWG , listed accessible boxes/covers/clamps, and whole-chain trip/reset/load testing.
- USB charging plan: 3 DC-fed USB PD station assemblies on separate 12V branches ( 1 office, 2 Galley), each standardized to 15A / 12 AWG . The third branch's actual fuse-panel slot/label is a field-audit item.

- AC-out-2 remains reserve-only in Phase 1 (labeled capped route; no energized branch hardware procured).

## Monitoring and Control Topology



## Fuse and Switch Housing Map (Where Each Item Is Physically Housed)

| Item ID | Item value/type                                                                  | Housing method                                 | Location                                                  |
|---------|----------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------|
| F-01A   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery A +                      |
| F-01B   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery B +                      |
| F-01C   | 200A Class T (provisional)                                                       | Blue Sea Class T fuse block (110A-200A family) | Battery compartment near Battery C +                      |
| F-02    | 125A MEGA                                                                        | Lynx integrated slot holder                    | Lynx Slot 1                                               |
| F-03    | 60A MEGA (80V Victron replacement stock)                                         | Lynx integrated slot holder                    | Lynx Slot 2                                               |
| F-04    | 200A/80V MEGA                                                                    | Lynx integrated slot holder                    | Lynx Slot 3 (dedicated alternator branch)                 |
| F-05    | 40A MEGA, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback | Lynx integrated slot holder                    | Lynx Slot 4; Orion 48v input feeder                       |
| F-06    | Not installed / retired standalone input-fuse position                           | No holder                                      | MIDI/FKS/DIN concepts superseded; do not stack after F-05 |

|                             |                                                                                                                              |                                                                                                                |                                                                                                                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| F-07                        | 60A MEGA (80V Victron replacement stock)                                                                                     | Victron MEGA fuse holder                                                                                       | Electrical cabinet at Orion 12V + source end                                                                                                                                 |
| F-09 PV string OCP position | 0 planned for the purchased single-string Renogy 4S candidate; no peer string can backfeed it                                | Add listed gPV hardware only if received labels, conductor/connector limits, or applicable PV rules require it | Roof-entry/service area if reopened                                                                                                                                          |
| F-10                        | Per branch (ATO/ATC)                                                                                                         | Integrated blade sockets in generic 12V fuse block                                                             | Electrical cabinet                                                                                                                                                           |
| AUDIO-HU / 12V-12           | 15A source/head-unit branch; KMC2 harness also contains a 15A ATM fuse                                                       | 12V fuse block branch plus KMC2 harness fuse                                                                   | Electrical cabinet to driver-side DC shelf/source face                                                                                                                       |
| AUDIO-SUB                   | 40A external fuse for Kicker PTRTP10 powered sub branch                                                                      | Inline/MRBF/AFS/ANL-class holder matched to selected 4 AWG kit; use 40A, not a generic 100A kit fuse           | Within about 18 in of the 12V source takeoff feeding the powered sub                                                                                                         |
| F-11                        | 100A class (12V buffer battery main)                                                                                         | Sealed inline MIDI/AMI/ANL holder                                                                              | Within ~`7"`` of 12V buffer battery positive post                                                                                                                            |
| F-12/F-13-PHVAN             | Littelfuse KLKD015.T, 15A/600VAC/DC fast-acting midget fuse                                                                  | Eaton/Bussmann HEB-AA 600V water-resistant in-line holder                                                      | Immediately beside the Lynx, fed from the F-04 alternator/load-side stud through short 14 AWG pigtails; sealed splice to 16 AWG PH-VAN red; no panel, DIN rail, or enclosure |
| CERB0-PWR                   | 1A-3A Cerbo GX power fuse                                                                                                    | Small inline holder rated for the 48v bank maximum                                                             | Electrical cabinet near 48v system-positive takeoff; system side of disconnect preferred for bench shutdown                                                                  |
| WS500 current-sense pair    | No fuse; common battery-negative shunt with purple/high on battery-negative side and grey/low on SmartShunt/Lynx system side | Twist pair if extended; route away from noise; configure Shunt at Battery                                      | Wakespeed shunt on the battery side of the hard-mounted SmartShunt                                                                                                           |
| F-15                        | 3A WS500 ignition/enable control fuse                                                                                        | Sealed inline ATC/ATO holder; 12V control circuit                                                              | Near Ford upfitter blunt-cut wire / WS500 control-wire handoff                                                                                                               |
| SW-12V-BATT                 | Manual battery disconnect switch                                                                                             | Sealed rotary DC switch body                                                                                   | Electrical cabinet near 12V fuse-block main + stud for service access                                                                                                        |
| OEM-SHUNT                   | External Victron-supplied inline fuse in red SmartShunt Vbatt+ cable                                                         | Inline holder in supplied red cable                                                                            | Prefer battery-side positive if SOC continuity is desired with the main disconnect open; system side is acceptable for zero disconnect-off parasitic draw                    |

Retired from active architecture:

- Obsolete pre-Mechman charger/fuse paths are removed from active board layout and labels.

## Conductor Schedule (Start-to-Finish)

| Segment ID | Circuit segment      | Nominal voltage | Current basis                | Overcurrent protection | Planned wire gauge | Estimate       |
|------------|----------------------|-----------------|------------------------------|------------------------|--------------------|----------------|
| C-01       | Battery A + -> F-01A | 48V             | Battery branch, fuse-limited | F-01A 200A provisional | 2/0 AWG            | 2.5 ft planned |
| C-02       | Battery B + -> F-01B | 48V             | Battery branch, fuse-limited | F-01B 200A provisional | 2/0 AWG            | 2.5 ft planned |
| C-02C      | Battery C + -> F-01C | 48V             | Battery branch, fuse-limited | F-01C 200A provisional | 2/0 AWG            | 2.5 ft planned |

|           |                                                                                                                                       |     |                                                                                                                  |                                        |                                                         |                    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------|--------------------|
| C-03      | Class T outputs -> battery-side 48V + busbar -> disconnect input                                                                      | 48V | Combined trunk current                                                                                           | F-01A/B/C                              | 2/0 AWG each branch                                     | 2.5 ft each branch |
| C-04      | Disconnect output -> Lynx + bus                                                                                                       | 48V | Aggregate discharge design current (165A = F-02 125A + Orion F-05 40A)                                           | Upstream Class T fuses                 | 2/0 AWG                                                 | 2.5 ft (ASSUMED)   |
| C-05      | Battery negatives -> battery-side 48V - busbar -> Wakespeed shunt battery/high side                                                   | 48V | Mixed-path rollup: 3x battery-negative branches at 82.5A design each + NEGBUS_TO_WSSHUNT trunk at 165A aggregate | N/A (main negative path)               | 2/0 AWG each branch                                     | 2.5 ft each branch |
| C-06      | Wakespeed shunt system/low side -> short 2/0 jumper -> SmartShunt battery side -> hard-mounted SmartShunt load side/Lynx - connection | 48V | Aggregate return current                                                                                         | N/A                                    | 2/0 AWG jumper; no separate SmartShunt-to-Lynx cable    | 0.5 ft jumper      |
| C-06A     | Lynx positive tap -> SmartShunt positive sense/power lead                                                                             | 48V | Shunt electronics supply (very low current)                                                                      | Factory inline fuse in OEM harness     | OEM harness lead                                        | 2.5 ft (ASSUMED)   |
| C-07      | Lynx Slot 1 (F-02) -> MultiPlus DC+                                                                                                   | 48V | Inverter branch, fuse-limited                                                                                    | F-02 125A                              | 2/0 AWG (manual minimum AWG 1 on short runs)            | 2.5 ft (ASSUMED)   |
| C-08      | MultiPlus DC- -> Lynx - bus                                                                                                           | 48V | Inverter return current                                                                                          | F-02 protects paired positive          | 2/0 AWG                                                 | 2.5 ft (ASSUMED)   |
| C-09      | MPPT BAT+ -> Lynx Slot 2 (F-03)                                                                                                       | 48V | Controller output (45A max)                                                                                      | F-03 60A/80V Victron row 188           | 6 AWG                                                   | 2.5 ft (ASSUMED)   |
| C-10      | MPPT BAT- -> Lynx - bus                                                                                                               | 48V | Controller return current                                                                                        | F-03 protects paired positive          | 6 AWG                                                   | 2.5 ft (ASSUMED)   |
| C-11      | Secondary alternator B+ -> APM-48 -> Lynx Slot 3 (F-04)                                                                               | 48V | Published alternator output reaches about 145.7A                                                                 | F-04 200A/80V                          | 2/0 AWG                                                 | 20 ft (ASSUMED)    |
| C-12      | Secondary alternator B- -> Lynx - bus (dedicated return)                                                                              | 48V | Alternator branch return current                                                                                 | F-04 paired                            | 2/0 AWG                                                 | 20 ft (ASSUMED)    |
| C-13/C-14 | Lynx Slot 4 (F-05) -> Orion 48V + input                                                                                               | 48V | Orion feeder, fuse-limited                                                                                       | F-05 40A MEGA, >=58VDC                 | Existing 6 AWG direct; 10 AWG adequate if ever replaced | 2.5 ft (ASSUMED)   |
| C-15      | Orion 48V - input -> Lynx - bus                                                                                                       | 48V | Orion input return current                                                                                       | Orion input positive protection paired | 6 AWG                                                   | 2.5 ft (ASSUMED)   |
| C-18      | Orion 12V + -> F-07 -> 12V fuse block                                                                                                 | 12V | Charger output path (30A continuous, 60A fuse)                                                                   | F-07 60A                               | 6 AWG planned (8 AWG minimum per Orion)                 | 2.5 ft (ASSUMED)   |



|       |                                                                                                                                                                                                      |                                                                    |                                                                                                                 |                                                                                                                |                                                                                                                                                                                                                                                                                   |                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
|       | main + stud                                                                                                                                                                                          |                                                                    |                                                                                                                 |                                                                                                                | table)                                                                                                                                                                                                                                                                            |                                                    |
| C-19  | Orion 12V - -> 12V fuse block integrated - bus / main - stud                                                                                                                                         | 12V                                                                | Charger output return                                                                                           | F-07 protects paired positive                                                                                  | 6 AWG                                                                                                                                                                                                                                                                             | 2.5 ft (ASSUMED)                                   |
| C-19A | 12V buffer battery + -> F-11 -> SW-12V-BATT -> 12V fuse block main + stud                                                                                                                            | 12V                                                                | Buffer source path and service isolation path                                                                   | F-11 100A class                                                                                                | 4 AWG planned                                                                                                                                                                                                                                                                     | 2.5 ft (ASSUMED)                                   |
| C-19B | 12V buffer battery - -> 12V fuse block integrated - bus / main - stud                                                                                                                                | 12V                                                                | Buffer battery return path                                                                                      | N/A (paired with C-19A)                                                                                        | 4 AWG planned                                                                                                                                                                                                                                                                     | 2.5 ft (ASSUMED)                                   |
| C-20  | 12V panel -> purchased Starlink DC converter                                                                                                                                                         | 12V                                                                | Purchased converter is the planned primary path; supplied AC power remains fallback until bench/load acceptance | Dedicated 20A ATO/ATC branch fuse planned; size remains limited by received input lead and installed conductor | Red/yellow marine duplex planned; 12 AWG only within about 6 ft one-way, otherwise 10 AWG                                                                                                                                                                                         | HOLD energy measured re-heat, and re-              |
| C-21  | 12V panel -> Fridge                                                                                                                                                                                  | 12V                                                                | Branch load                                                                                                     | F-10 15A                                                                                                       | 12 AWG duplex                                                                                                                                                                                                                                                                     | 12 ft (ASSUMED prior voltage)                      |
| C-22  | 12V panel -> LF Bros diesel-heater harness                                                                                                                                                           | 12V                                                                | Startup/cooldown branch; keep energized through controller-commanded shutdown                                   | F-10 15A (retain any supplied harness fuse; verify coordination)                                               | 12 AWG duplex                                                                                                                                                                                                                                                                     | 8 ft (ASSUMED)                                     |
| C-23  | 12V panel -> Water pump                                                                                                                                                                              | 12V                                                                | Branch load                                                                                                     | F-10 10A                                                                                                       | 14 AWG duplex                                                                                                                                                                                                                                                                     | 8 ft (ASSUMED)                                     |
| C-24  | 12V panel -> CO + propane detector                                                                                                                                                                   | 12V                                                                | Branch load                                                                                                     | F-10 3A                                                                                                        | 18/2                                                                                                                                                                                                                                                                              | 8 ft (ASSUMED)                                     |
| C-25  | 12V panel -> LED lights + dimmer (Hiatus pre-installed)                                                                                                                                              | 12V                                                                | Branch load                                                                                                     | F-10 5A                                                                                                        | 18/2                                                                                                                                                                                                                                                                              | 8 ft (ASSUMED)                                     |
| C-26  | 48V system positive/negative -> CERBO-PWR -> Cerbo GX power input                                                                                                                                    | 48V                                                                | Cerbo electronics feed (~3W)                                                                                    | CERBO-PWR 1A-3A inline fuse                                                                                    | 18 AWG red/black duplex acceptable                                                                                                                                                                                                                                                | 2.5 ft (ASSUMED)                                   |
| C-27  | Purchased Renogy 4S string -> two-polarity roof transition box -> continuous front-gap retractile -> fixed-shell premium gland -> two accessible sealed splices -> stationary 12/2 -> two-pole load- | 78.0V Vmp; 95.6V Voc; 114.86V nominal-coefficient cold Voc at -40C | 8.98A Imp; 9.50A Isc STC; 14.8A conservative conductor screen                                                   | No F-09 presently planned for one string; reopen only from received-label/code review                          | Preferred 12 AWG x 2 fixed/moving PV path after final route/derating proof; moving cord requires written >=150VDC, coiled ampacity, outdoor/wet/UV/cold/flex proof. Current envelope is ~24 in down / ~52 in up; research 60-66 in rated overall. No four-to-one branch/combiner. | No exterior lengths/OD/ remain on fu restart/track |

|      |                                                                                                                    |          |                                                                                |                                                                                    |                                                            |                                |
|------|--------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------|
|      | break -> MPPT PV input                                                                                             |          |                                                                                |                                                                                    |                                                            |                                |
| C-28 | Shore source/adapters -> portable EMS -> shore cord -> shore inlet -> combined AC DIN enclosure / AC input breaker | 120VAC   | Source-limited shore current (adapter-constrained at source)                   | 30A AC input breaker/disconnect baseline with source-current-limit settings policy | 10/3 shore feed to inlet/AC-in area                        | 8 ft (ASSUMED)                 |
| C-29 | AC input breaker/disconnect -> MultiPlus AC-in                                                                     | 120VAC   | MultiPlus AC input current (30A hardware basis)                                | Upstream 30A AC breaker/disconnect (C-28)                                          | 10 AWG stranded AC conductors                              | 2.5 ft (ASSUMED)               |
| C-30 | MultiPlus AC-out-1 -> combined AC DIN enclosure / AC-out main breaker                                              | 120VAC   | Inverter-backed AC-out feeder current (30A system cap)                         | 30A AC-out main breaker                                                            | 10/3 stranded AC feeder                                    | 2.5 ft (ASSUMED)               |
| C-31 | Branch A -> office/driver GFCI receptacle                                                                          | 120VAC   | Branch load (office monitor/chargers/general outlet use)                       | 20A branch breaker + GFCI receptacle                                               | 12/3 stranded AC branch cable                              | 15 ft (ASSUMED)                |
| C-32 | Branch B -> galley/passenger GFCI receptacle                                                                       | 120VAC   | Branch load (galley/general high-draw outlet; induction/Ninja SP151 sequenced) | 20A branch breaker + GFCI receptacle                                               | 12/3 stranded AC branch cable                              | 15 ft (ASSUMED)                |
| C-33 | MultiPlus AC-out-2 (reserve-only) -> capped route for future shore-only branch                                     | 120VAC   | N/A in Phase 1 (route reserved only)                                           | N/A in Phase 1 (no energized branch hardware)                                      | 12 AWG stranded AC conductors (reserve path only)          | 15 ft (ASSUMED)                |
| C-34 | 12V panel -> USB PD station branch (office zone)                                                                   | 12V      | One Acegoo 118W station                                                        | F-10 branch fuse (15A)                                                             | 12 AWG duplex baseline                                     | 5 ft (ASSUMED)                 |
| C-35 | 12V panel -> USB PD station branch (galley zone)                                                                   | 12V      | Galley charging branch (65W class USB-C plus USB-A/C loads)                    | F-10 branch fuse (15A)                                                             | 12 AWG duplex baseline                                     | 8 ft (ASSUMED)<br>rough-in for |
| C-36 | 12V panel -> Maxxair fan (Hiatus pre-installed)                                                                    | 12V      | Roof ventilation branch                                                        | F-10 branch fuse (10A)                                                             | 14 AWG duplex baseline                                     | 8 ft (ASSUMED)                 |
| C-37 | 12V panel -> DC ambient/cabinet LED strips (planned Govee)                                                         | 12V      | Branch load                                                                    | F-10 branch fuse (5A)                                                              | 18/2 baseline                                              | 8 ft (ASSUMED)                 |
| C-38 | F-04 alternator/load-side stud -> Eaton/Bussmann HEB-AA + Littelfuse KLKD015.T ->                                  | 48V bank | Combined regulator-power / positive-voltage-sense feed                         | F-12/F-13-PHVAN (15A)                                                              | Short 14 AWG pigtails, sealed transition to 16 AWG harness | In-line beside                 |

|         |                                                                                                                                    |                                               |                                                              |                                                             |                                                                              |                          |
|---------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------|
|         | WS500 PH-VAN red lead                                                                                                              |                                               |                                                              |                                                             |                                                                              |                          |
| C-39    | Retired separate WS500 positive-sense conductor                                                                                    | N/A                                           | Not installed with confirmed PH-VAN harness                  | None                                                        | N/A                                                                          | N/A                      |
| C-40    | WS500 current-sense pair to common battery-negative shunt: purple/high battery-negative side, grey/low SmartShunt/Lynx system side | low-current sense                             | Net battery-current feedback                                 | No fuse; twist pair if extended; configure Shunt at Battery | Harness lead                                                                 | Local near h             |
| C-41    | Ford Upfitter #3 output -> F-15 -> WS500 brown ignition/enable wire                                                                | 12V control lead                              | Manual alternator-enable signal only                         | F-15 (3A)                                                   | 16 AWG TXL/GXL                                                               | 6 ft (ASSUMED            |
| C-42    | 12V panel -> Kicker 46KMC2 media center/source unit                                                                                | 12V                                           | Source/head-unit branch (15A max; KMC2 manual shows 15A ATM) | F-10/AUDIO-HU 15A branch plus KMC2 harness fuse             | 12 AWG duplex if kept near 5 ft; use 10 AWG if longer                        | 5 ft (ASSUMED            |
| C-43    | 12V source/main + stud -> AUDIO-SUB source fuse -> Kicker 49PTRTP10 powered sub +                                                  | 12V                                           | Powered sub branch; PTRTP10 manual external fuse value       | AUDIO-SUB 40A                                               | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED junction)  |
| C-44    | Kicker 49PTRTP10 powered sub - -> 12V negative bus/main - stud                                                                     | 12V                                           | Powered sub return                                           | AUDIO-SUB protects paired positive                          | 4 AWG tinned/OFC                                                             | 8 ft (ASSUMED            |
| C-45    | KMC2 remote turn-on output -> PTRTP10 remote input                                                                                 | 12V low-current control                       | Remote amp/sub enable                                        | KMC2/source branch protected                                | 18 AWG                                                                       | 8 ft (ASSUMED            |
| C-46    | KMC2 RCA line-out -> PTRTP10 RCA input                                                                                             | low-level audio signal                        | Subwoofer signal                                             | N/A                                                         | shielded 2-channel RCA                                                       | measured ro              |
| C-47L/R | KMC2 front speaker outputs - > left/right Kicker CSC67 speakers                                                                    | speaker-level audio                           | One 4 ohm speaker per KMC2 channel                           | KMC2 internal protection / AUDIO-HU source branch           | 16 AWG marine speaker wire                                                   | 10-12 ft eac             |
| C-48    | KUS SSS/SSL sender black signal + pink return -> matching signal/negative pair in Cerbo GX MK2 Tank 1 column                       | Cerbo low-current resistive-sender excitation | Fresh-water level, US 240-30 ohm                             | No external fuse or power feed; do not share chassis return | 18-22 AWG tinned duplex, ferrules at Cerbo, sealed pigtail splices at sender | Measure aft (ASSUMED <10 |

|      |                                                 |     |                                |                                                                            |               |                              |
|------|-------------------------------------------------|-----|--------------------------------|----------------------------------------------------------------------------|---------------|------------------------------|
| C-49 | 12V panel -><br>second Galley USB<br>PD station | 12V | Third independent PD<br>branch | F-10 branch fuse<br>(15A); exact panel<br>slot/label to verify<br>in field | 12 AWG duplex | Owner report<br>final length |
|------|-------------------------------------------------|-----|--------------------------------|----------------------------------------------------------------------------|---------------|------------------------------|

## Wiring Validation Worksheet (Estimate Pass, 2026-02-18)

Calculation basis for drop screening:

- $V_{drop} = I * (2 * L_{one\_way} * R_{per\_ft})$
- Resistance basis used in this pass ( ohm/ft ): 2/0=0.0000779 , 6 AWG=0.0003951 , 4 AWG=0.0002485 , 12 AWG=0.001588 , 14 AWG=0.002525 , 18/2=0.006385 , 10 AWG=0.000999 .
- Design targets: <=2% on major 48V trunks, <=3% on planned 12V /AC branches.
- C-05 is a rollup row that includes both branch and trunk return paths; voltage-drop screen shown is the conservative interim worst-case ( 155A ) within that rollup.

| Circuit ID | From               | To                           | Fuse       | Current basis                     | Gauge   | Estimated one-way length                                                                                  | Voltage drop %                      |
|------------|--------------------|------------------------------|------------|-----------------------------------|---------|-----------------------------------------------------------------------------------------------------------|-------------------------------------|
| C-01       | Battery A +        | F-01A                        | F-01A 200A | 77.5A interim design branch share | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder |
| C-02       | Battery B +        | F-01B                        | F-01B 200A | 77.5A interim design branch share | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder |
| C-02C      | Battery C +        | F-01C                        | F-01C 200A | 77.5A interim design branch share | 2/0 AWG | 2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced | 0.059% @ 51.2V planning placeholder |
| C-03       | Class T load studs | 48V + bus / disconnect input | F-01A/B/C  | 77.5A interim per branch          | 2/0 AWG | 2.5 ft each (x4 conductors)                                                                               | 0.059% @ 51.2V                      |

|           |                                 |                                                                              |                                        |                                                                                                        |                           |                             |                                   |
|-----------|---------------------------------|------------------------------------------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------|-----------------------------------|
| C-04      | Disconnect output               | Lynx + bus                                                                   | Upstream Class T                       | 155A interim / 145A final aggregate                                                                    | 2/0 AWG                   | 2.5 ft                      | 0.12% @ 51.2V interim             |
| C-05      | Battery - branches              | Wakespeed shunt battery/high side via 48V - bus                              | N/A                                    | 77.5A per battery-negative branch; row rollup also includes one 155A interim trunk (NEGBUS_TO_WSSHUNT) | 2/0 AWG                   | 2.5 ft each (x4 conductors) | 0.12% @ 51.2V (worst-case rollup) |
| C-06      | Wakespeed shunt system/low side | SmartShunt battery side; SmartShunt load side remains hard-mounted to Lynx - | N/A                                    | 155A interim / 145A final aggregate return                                                             | Short 2/0 AWG jumper      | 0.5 ft placeholder          | 0.024% @ 51.2V interim            |
| C-06A     | Lynx positive tap               | SmartShunt sense/power lead                                                  | OEM inline fuse                        | OEM harness current                                                                                    | OEM harness               | 2.5 ft                      | N/A (low-current OEM lead)        |
| C-07      | Lynx Slot 1 DC+                 | MultiPlus DC+                                                                | F-02 125A                              | 125A                                                                                                   | 2/0 AWG                   | 2.5 ft                      | 0.10% @ 51.2V                     |
| C-08      | MultiPlus DC-                   | Lynx - bus                                                                   | F-02 paired                            | 125A                                                                                                   | 2/0 AWG                   | 2.5 ft                      | 0.10% @ 51.2V                     |
| C-09      | MPPT BAT+                       | Lynx Slot 2                                                                  | F-03 60A (80V Victron row 188)         | 45A                                                                                                    | 6 AWG                     | 2.5 ft                      | 0.17% @ 51.2V                     |
| C-10      | MPPT BAT-                       | Lynx - bus                                                                   | F-03 paired                            | 45A                                                                                                    | 6 AWG                     | 2.5 ft                      | 0.17% @ 51.2V                     |
| C-11      | Secondary alternator B+         | Lynx Slot 3 via APM-48                                                       | F-04 200A/80V                          | 145.7A published-curve screen                                                                          | 2/0 AWG                   | 20 ft                       | <0.80% @ 58.4V                    |
| C-12      | Secondary alternator B-         | Lynx - bus (dedicated return)                                                | F-04 paired                            | 145.7A published-curve screen                                                                          | 2/0 AWG                   | 20 ft                       | <0.80% @ 58.4V                    |
| C-13/C-14 | Lynx Slot 4                     | Orion 48V +                                                                  | F-05 40A MEGA, >=58VDC                 | 40A fuse basis                                                                                         | Existing 6 AWG direct     | 2.5 ft                      | 0.16% @ 51.2V                     |
| C-15      | Orion 48V -                     | Lynx - bus                                                                   | Orion input positive protection paired | 40A fuse basis                                                                                         | 6 AWG                     | 2.5 ft                      | 0.16% @ 51.2V                     |
| C-18      | Orion 12V +                     | Fuse block main + stud                                                       | F-07 60A/80V Victron row 188           | 30A                                                                                                    | 6 AWG                     | 2.5 ft                      | 0.49% @ 12V                       |
| C-19      | Orion 12V -                     | Fuse block integrated - bus / main - stud                                    | F-07 paired                            | 30A                                                                                                    | 6 AWG                     | 2.5 ft                      | 0.49% @ 12V                       |
| C-19A     | Buffer battery +                | Fuse block main + stud (via F-11/SW)                                         | F-11 100A                              | 50A design                                                                                             | 4 AWG                     | 2.5 ft                      | 0.52% @ 12V                       |
| C-19B     | Buffer battery -                | Fuse block integrated - bus / main - stud                                    | N/A                                    | 50A design                                                                                             | 4 AWG                     | 2.5 ft                      | 0.52% @ 12V                       |
| C-20      | 12V fuse panel                  | Purchased Starlink DC converter                                              | Dedicated 20A ATO/ATC                  | Measure received input/current                                                                         | 12 AWG marine duplex only | Measure                     | Recalculate from                  |

|      |                                                                                                                                                                             |                                                  | branch<br>planned; not<br>installed                  | behavior                                                                 | within about 6<br>ft one-way;<br>otherwise 10<br>AWG     |                                                   | measured<br>route/current                    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------|----------------------------------------------|
| C-21 | 12V fuse panel                                                                                                                                                              | Fridge                                           | F-10 15A                                             | 7A                                                                       | 12 AWG duplex                                            | 12 ft                                             | 2.23% @ 12V                                  |
| C-22 | 12V fuse panel                                                                                                                                                              | LF Bros diesel-<br>heater harness                | F-10 15A +<br>supplied<br>harness fuse<br>if present | 10A startup screen                                                       | 12 AWG duplex                                            | 8 ft<br>assumed                                   | 2.12% @ 12V                                  |
| C-23 | 12V fuse panel                                                                                                                                                              | Water pump                                       | F-10 10A                                             | 7A                                                                       | 14 AWG duplex                                            | 8 ft                                              | 2.36% @ 12V                                  |
| C-24 | 12V fuse panel                                                                                                                                                              | CO+propane<br>detector                           | F-10 3A                                              | 0.2A                                                                     | 18/2                                                     | 8 ft                                              | 0.17% @ 12V                                  |
| C-25 | 12V fuse panel                                                                                                                                                              | LED lights +<br>dimmer (Hiatus<br>pre-installed) | F-10 5A                                              | 5A                                                                       | 18/2                                                     | 8 ft                                              | 4.26% @ 12V                                  |
| C-26 | 48V system feed                                                                                                                                                             | Cerbo GX                                         | CERB0-PWR 1A-<br>3A                                  | ~3W                                                                      | 18 AWG duplex<br>acceptable                              | 2.5 ft                                            | N/A (low-<br>current<br>electronics<br>feed) |
| C-27 | Purchased Renogy<br>4S string + two-<br>polarity transition<br>box + continuous<br>front-gap retractile<br>+ premium gland<br>+ stationary 12/2<br>+ two-pole<br>disconnect | MPPT PV input                                    | No F-09<br>presently<br>planned                      | 9.50A I <sub>sc</sub> STC /<br>14.8A<br>conservative<br>conductor screen | Preferred 12<br>AWG x 2 after<br>route/derating<br>proof | Full route<br>pending; 20<br>ft one-way<br>screen | 0.94% at 9.5A<br>/ 78V for 12<br>AWG         |
| C-28 | Shore inlet path                                                                                                                                                            | AC input<br>breaker/disconnect                   | Source-<br>limited AC<br>OCP                         | 30A hardware<br>basis                                                    | 10/3                                                     | 8 ft                                              | 0.40% @<br>120VAC                            |
| C-29 | AC input<br>breaker/disconnect                                                                                                                                              | MultiPlus AC-in                                  | Upstream<br>AC OCP                                   | 30A hardware<br>basis                                                    | 10 AWG AC                                                | 2.5 ft                                            | 0.12% @<br>120VAC                            |
| C-30 | MultiPlus AC-out-1                                                                                                                                                          | AC-out main<br>breaker                           | 30A AC-out<br>main                                   | 30A hardware<br>basis                                                    | 10/3                                                     | 2.5 ft                                            | 0.12% @<br>120VAC                            |
| C-31 | Branch A                                                                                                                                                                    | Office/driver<br>receptacle chain                | 20A branch<br>OCP                                    | 20A                                                                      | 12 AWG AC                                                | 15 ft                                             | 0.79% @<br>120VAC                            |

|           |                                                                                                                  |                                               |                                                    |                                                                   |                                   |                       |                                    |
|-----------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------|-----------------------------------|-----------------------|------------------------------------|
| C-32      | Branch B                                                                                                         | Galley/passenger receptacle chain             | 20A branch OCP                                     | 20A                                                               | 12 AWG AC                         | 15 ft                 | 0.79% @ 120VAC                     |
| C-33      | MultiPlus AC-out-2                                                                                               | Reserve-only capped route                     | N/A in Phase 1                                     | N/A                                                               | 12 AWG AC (reserve path only)     | 15 ft                 | 0.60% @ 120VAC (future-use screen) |
| C-34      | 12V fuse panel                                                                                                   | Office USB PD station                         | F-10 15A                                           | 15A design cap                                                    | 12 AWG duplex                     | 5 ft                  | 1.99% @ 12V                        |
| C-35      | 12V fuse panel                                                                                                   | Galley USB PD station                         | F-10 15A                                           | 8A expected                                                       | 12 AWG duplex                     | 8 ft                  | 1.69% @ 12V                        |
| C-49      | 12V fuse panel                                                                                                   | Second Galley USB PD station                  | F-10 15A; slot to verify                           | 15A design cap                                                    | 12 AWG duplex                     | Field routed; measure | Recalculate from measured length   |
| C-36      | 12V fuse panel                                                                                                   | Maxxair fan (Hiatus pre-installed)            | F-10 10A                                           | 4A expected                                                       | 14 AWG duplex                     | 8 ft                  | 1.35% @ 12V                        |
| C-37      | 12V fuse panel                                                                                                   | DC ambient/cabinet LED strips (planned Govee) | F-10 5A                                            | 5A design cap                                                     | 18/2                              | 8 ft                  | 4.26% @ 12V                        |
| C-38      | F-04 alternator/load-side stud                                                                                   | WS500 PH-VAN short red lead                   | F-12/F-13-PHVAN Littelfuse KLKD015.T 15A/600VAC/DC | combined regulator-power / voltage-sense feed at 48V bank voltage | 14 AWG pigtails to 16 AWG harness | Local                 | N/A (harness-limited)              |
| C-39      | Retired separate positive-sense source                                                                           | Not installed                                 | None                                               | superseded by confirmed PH-VAN combined lead                      | N/A                               | N/A                   | N/A                                |
| C-40      | Common battery-negative Wakespeed shunt: purple/high battery-negative side, grey/low SmartShunt/Lynx system side | WS500 current-sense input                     | No fuse                                            | low-current sense; twist pair if extended                         | Harness lead                      | Local                 | N/A (harness-limited)              |
| C-41      | Ford Upfitter #3 output                                                                                          | WS500 brown ignition/enable input via F-15    | F-15 3A                                            | manual low-current control only                                   | 16 AWG TXL/GXL                    | 6 ft                  | N/A (control circuit)              |
| C-42      | 12V panel                                                                                                        | Kicker 46KMC2 media center                    | AUDIO0-HU 15A branch + KMC2 15A ATM harness fuse   | 15A max fuse basis                                                | 12 AWG duplex                     | 5 ft                  | 1.99% @ 12V                        |
| C-43/C-44 | 12V source/main studs                                                                                            | Kicker 49PTRTP10 powered sub                  | AUDIO0-SUB 40A external source fuse                | 40A manual fuse basis                                             | 4 AWG positive + matching return  | 8 ft                  | 1.33% @ 12V                        |
| C-45      | KMC2 remote output                                                                                               | PTRTP10 remote input                          | KMC2/source branch                                 | low-current remote                                                | 18 AWG                            | 8 ft                  | N/A                                |

|         |                            |                           |                              |                               |                            |                |     |
|---------|----------------------------|---------------------------|------------------------------|-------------------------------|----------------------------|----------------|-----|
|         |                            |                           | protected                    |                               |                            |                |     |
| C-46    | KMC2 RCA line-out          | PTRTP10 RCA input         | N/A                          | low-level signal              | shielded RCA               | measured route | N/A |
| C-47L/R | KMC2 front speaker outputs | Left/right CSC67 speakers | KMC2/source branch protected | one 4 ohm speaker per channel | 16 AWG marine speaker wire | 10-12 ft each  | N/A |

## Wire Rollup (No-Padding Purchase Baseline)

| Gauge / cable family                       | Estimated total                                                                                                                      | Source circuits                                                                                                                                                                                                                                           | BOM row                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2/0 AWG red                                | 42.5 ft                                                                                                                              | C-01, C-02, C-02C, C-03, C-04, C-07, C-11                                                                                                                                                                                                                 | 28                     |
| 2/0 AWG black                              | 33.0 ft                                                                                                                              | C-05, C-06, C-08, C-12                                                                                                                                                                                                                                    | 28                     |
| 6 AWG red                                  | 7.5 ft                                                                                                                               | C-09, C-13/C-14, C-18                                                                                                                                                                                                                                     | 29                     |
| 6 AWG black                                | 7.5 ft                                                                                                                               | C-10, C-15, C-19                                                                                                                                                                                                                                          | 29                     |
| 4 AWG red                                  | 10.5 ft (2.5 ft existing + 8 ft audio sub planning branch)                                                                           | C-19A, C-43                                                                                                                                                                                                                                               | 30, 192                |
| 4 AWG black                                | 10.5 ft (2.5 ft existing + 8 ft audio sub return)                                                                                    | C-19B, C-44                                                                                                                                                                                                                                               | 30, 192                |
| 12 AWG PV cable allowance                  | Route length pending                                                                                                                 | C-27 purchased Renogy 4S; three series mates, two-polarity roof transition, continuous front-gap wet/flex retractile, stationary straight lower gland tail, then two sealed splices to stationary 12/2; no four-to-one combiner or F-09 presently planned | 31, 106, 121           |
| 14 AWG duplex                              | 16 ft                                                                                                                                | C-23, C-36                                                                                                                                                                                                                                                | 32                     |
| 18/2                                       | 24.0 ft plus separate Cerbo low-current duplex                                                                                       | C-24, C-25, C-37; Cerbo C-26 uses 48V fused low-current duplex                                                                                                                                                                                            | 33 / low-current stock |
| 12 AWG AC branch cable                     | 30 ft (C-33 excluded in Phase 1)                                                                                                     | C-31, C-32                                                                                                                                                                                                                                                | 113                    |
| 10/3 shore + AC-in/out feed                | 13 ft                                                                                                                                | C-28, C-29, C-30                                                                                                                                                                                                                                          | 114                    |
| 12 AWG fridge/heater/USB/audio branch mix  | Fridge: 12 ft; heater: 8 ft; USB: 5 ft office + 8 ft Galley A + measured Galley B field run; audio source: 5 ft short-run assumption | C-21, C-22, C-34, C-35, C-49, C-42                                                                                                                                                                                                                        | 116, 193 / field stock |
| Camper audio signal/speaker/control wiring | RCA measured route, 18 AWG remote, and 16 AWG marine speaker wire runs                                                               | C-45, C-46, C-47L/R                                                                                                                                                                                                                                       | 193                    |
| WS500 harness/sense leads                  | Included in selected kit/harness set; C-39 is retired under PH-VAN                                                                   | C-38, C-40                                                                                                                                                                                                                                                | 168, 171               |
| 16 AWG TXL/GXL control wire                | 6 ft                                                                                                                                 | C-41                                                                                                                                                                                                                                                      | 176                    |

Notes:

- This table is a base estimate only; it intentionally excludes order padding and termination waste.
- Apply personal order overage at checkout based on actual spool cut increments and routing confidence.
- Parallel bank balancing is locked by similar total loop resistance per battery. Positive-only leads may differ if the paired negative path offsets the difference; verify with final measured lengths and, if needed, clamp-current checks.



# 3x Battery Bank Bench-Build Cut List (2/0 AWG)

Purpose: make the bench build orderable without needing final camper run lengths. Treat lengths below as *bench module* lengths only; final install harnesses should be re-cut after layout freeze.

Assumptions:

- Battery terminals are M8 (verify your battery stud size before ordering lugs).
- Class T fuse blocks and battery-side busbars use 3/8" studs (treat as M10 lugs unless your specific hardware differs).
- Lynx Distributor is the M10 model (main connections M10 ; internal/fuse studs may still be M8 depending on the position).

| Cable ID              | Qty     | From -> To                                                         | Color | Gauge | Est. one-way length      | Lug A             | Lug B             |
|-----------------------|---------|--------------------------------------------------------------------|-------|-------|--------------------------|-------------------|-------------------|
| BATT+_A/B/C           | 3       | Battery + -> Class T block line side                               | red   | 2/0   | 2.5 ft each (ASSUMED)    | M8                | M10               |
| FUSE_TO_POSBUS_A/B/C  | 3       | Class T block load side -> 48V + busbar                            | red   | 2/0   | 2.5 ft each (ASSUMED)    | M10               | M10               |
| POSBUS_TO_DISC        | 1       | 48V + busbar -> disconnect input                                   | red   | 2/0   | 2.5 ft (ASSUMED)         | M10               | M10               |
| DISC_TO_LYNX+         | 1       | disconnect output -> Lynx + input                                  | red   | 2/0   | 2.5 ft (ASSUMED)         | M10               | M10               |
| BATT-_A/B/C           | 3       | Battery - -> 48V - busbar                                          | black | 2/0   | 2.5 ft each (ASSUMED)    | M8                | M10               |
| NEGBUS_TO_WSSHUNT     | 1       | 48V - busbar -> Wakespeed shunt battery/high side                  | black | 2/0   | 2.5 ft (ASSUMED)         | M10               | Verify shunt stud |
| WSSHUNT_TO_SMARTSHUNT | 1       | Wakespeed shunt system/low side -> SmartShunt battery side         | black | 2/0   | 0.5 ft placeholder       | Verify shunt stud | M10               |
| SMARTSHUNT_TO_LYNX-   | 0 cable | SmartShunt load side remains hard-mounted directly to Lynx - input | —     | —     | Existing stud connection | M10               | M10               |
| LYNX_SLOT1_TO_MULTI+  | 1       | Lynx Slot 1 DC+ -> MultiPlus DC+                                   | red   | 2/0   | 2.5 ft (ASSUMED)         | M8                | M8                |
| LYNX_TO_MULTI-        | 1       | Lynx - -> MultiPlus DC-                                            | black | 2/0   | 2.5 ft (ASSUMED)         | M8                | M8                |

Locked balancing rule for the 3x parallel bank:

- Keep each battery path's total positive + negative loop resistance similar.
- Equal positive-only leads are not required when the negative lead lengths intentionally offset the positive lead differences.
- Keep the same cable family, lug geometry, fuse/holder family, and termination quality across all three battery paths.
- Verify sharing with clamp-current checks under charge/load if final measured path lengths differ materially.

Torque reference (verify against your exact manuals/hardware):

- MultiPlus-II DC terminals: 12 Nm ( M8 nut) per Victron installation guidance.
- SmartShunt shunt bolts: verify torque for the installed 300A model per Victron installation guidance.
- Lynx Distributor M10 model: M10 nuts 33 Nm (older serials may be lower), and M8 nuts 14 Nm per Victron Lynx installation guidance.

## Additional Components Included In Topology Scope

- 48V disconnect ( 275A )
- Pre-charge resistor (commissioning/soft-charge aid before connecting large DC loads)
- Battery-side 48V + combine busbar (after Class T fuses)
- Battery-side 48V - combine busbar (battery-only, before SmartShunt)
- 12V fuse block main + stud used as source-combine point (Orion + buffer battery feed)
- 12V fuse block integrated negative bus/main - used as shared return point
- 12V buffer battery main fuse ( F-11 ) and manual disconnect switch ( SW-12V-BATT )
- Shore AC inlet + cord/adapter interface hardware

- Portable EMS in source-side shore path before camper inlet
- Single combined 6-way AC DIN enclosure with isolated AC-in and AC-out neutral paths plus common equipment grounding/PE handling
- AC-out 30A main breaker plus 20A / 20A branch breaker and GFCI receptacle hardware
- Receptacle boxes + 120V outlets (current purchased baseline 2 first-in-chain GFCI receptacles/covers in row 15 ; active row 112 now records six purchased old-work boxes for field-fit downstream devices, but device count/covers/clamps/fill/access/testing remain open)
- AC-out-2 reserve-only capped route (no energized Phase 1 branch hardware)
- USB PD station branch hardware ( 3 stations / independent routes owner-reported 2026-08-05 : 1x office and 2x Galley; row 115 owns the exact purchased Acegoo evidence while the additional station purchase provenance remains untracked)
- Camper audio hardware: Kicker 46KMC2 source branch, Kicker 49PTRTP10 powered sub branch, AUDIO-HU 15A source protection, AUDIO-SUB 40A source fuse, RCA/speaker/remote wiring
- Battery temperature sensor wiring to inverter/monitoring path
- SmartShunt fused positive sense/power lead (factory harness)
- Cerbo GX CERB0-PWR small inline fused 48V power feed
- Ford Upfitter #3 control lead and local F-15 inline fuse for the WS500 brown ignition/enable wire

## Assumptions (Explicit)

1. Cable sizing assumes fine-strand copper conductors (OFC welding-cable baseline for high-current DC paths), enclosed vehicle routing, and the estimated one-way lengths listed in this document.
2. Voltage-drop design intent used here:  $\leq 2\%$  on major 48V power runs and  $\leq 3\%$  on 12V branch circuits.
3. The purchased Renogy panels are currently screened as one 4S string and presently use no F-09 series fuse because no peer string can backfeed them. Received labels must confirm Isc and maximum series-fuse rating; reopen OCP if conductor/connector limits or applicable PV rules require it.
4. Cerbo GX feed is now a small inline fused 48V feed ( CERB0-PWR ) from the system/load side of the main disconnect during bench commissioning, so the Cerbo powers down with the house system.
5. Orion branch uses one source-side fuse only: F-05 40A MEGA, body-marked at least 58VDC , in Lynx Slot 4 feeding the existing 6 AWG input pair directly. The 58VDC minimum is tied to the locked 56.8V charge ceiling; use Victron CIP138040020 40A/80V if replacement is needed. Standalone F-06 is retired; do not add an inline fuse after the Lynx slot.
6. No low-voltage-disconnect (LVD) automation is included in Phase 1; protection is source fusing plus manual SW-12V-BATT isolation.
7. Alternator architecture lock is dedicated 48V secondary alternator path ( Mechman + WS500 + APM-48 ) with F-04 200A/80V ; obsolete pre-Mechman engine-bay fuse paths are removed from active layout.
8. F-01A/B/C are provisionally set to 200A pending final 51.2V battery datasheet/manual confirmation; if validated limits are lower, shift to 175A .
9. 2/0 cable quantity planning baseline in this pass is 77.5 ft total no-padding ( 42.5 ft red + 35.0 ft black); user-applied order padding is intentionally deferred to checkout.
10. Manual alternator shutdown baseline is Ford Upfitter #3 feeding the WS500 brown ignition/enable wire through F-15 ; WS500 white Feature-In remains a future-only reserve for automatic interlock work.
11. Camper audio is a 12V accessory load. KMC2 source branch is 15A ; PTRTP10 sub branch is 40A with 4 AWG power and return. Audio peaks can exceed Orion continuous output and rely on the 12V buffer battery, so first loud testing should watch 12V voltage/SOC.

## Completion Status

- DC/PV topology is complete for current BOM scope and load model scope.
- AC hierarchy is complete at architecture level, including transfer behavior, branch strategy, and protection chain.
- Full-circuit estimate pass is now documented with run lengths, voltage-drop screening, and purchase rollups.
- Camper audio branch added as a DC-first 12V accessory package: Kicker 46KMC2 source branch plus Kicker 49PTRTP10 powered sub branch; detailed product/routing notes live in docs/implementation/CAMPER\_audio\_system.md .
- Remaining work is final measured-length replacement and SKU-level closeout before final cut-to-length harness production.